Moving the sun in architecture photos

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The short version

I wanted to change where the sun comes from in a photo without changing the building. Nothing I tried does this perfectly. The best tool for outdoor shots was the Sun-Direction LoRA: it really moves the sun, and on the villa you can clearly see front, back, top and sunset light. But it still changes some materials, and indoors it changes the furniture. So I would launch it for exteriors only, with an automatic check in front of it.

What I used and why

Tool	Why I picked it	How I ran it
Sun-Direction LoRA (9B, then 4B)	The only image tool I found made just for moving the sun	Free Hugging Face Space, then on my own RTX 4070
Qwen 3D Lighting	You pick a light direction, and it can try interiors	Free Hugging Face Space
FLUX.1 Kontext	A normal text-edit model, to see how far a plain prompt gets	Free Hugging Face Space
Light transfer (my own script)	To keep the original pixels and only copy the new light	Python, on my own computer

How I used them

- I took three real photos from Unsplash: a villa, a concrete apartment block and a living room.
- I asked each tool to move the sun, then put every result next to the original.
- I wrote small Python scripts to send the photos, save the results and compare them.
- To judge a result I asked three things: did the light really change, did the building stay the same, and did single objects like the sofa stay the same? A score helped me sort results, but the final call was by eye.

Time and cost per image

Tool	Time	Cost for me
Sun-Direction 9B (Space)	20 to 26 s	$0
Sun-Direction 4B (my RTX 4070)	73 to 116 s	$0
Qwen 3D Lighting (Space)	6.5 to 17 s	$0
FLUX.1 Kontext (Space)	31 to 45 s	$0

Space times include waiting in the queue. My 4B runs were slow because the model didn't fully fit on my 12 GB card.

The villa: sun from different sides

This is the best result of the project. Same photo, same model, only the sun position changes. The reflections on the pool and the shadows on the walls change with it.



Original photo



Sun behind the camera
Flat, even light on the front of the house.



Sun behind the building
You can see the sun in the sky, and the walls fall into shade.



Sun straight above
Short, hard shadows under the overhangs.



Low sun from the left
Lower sun gives longer shadows across the terrace.



Low sun from the right
Now the right wall lights up. Left and right are still the closest pair.



Higher sun from the left
My first try at this angle. Softer than the low sun.



Higher sun from the right
Same here. Left and right looked too similar, so I lowered the sun.



Sunset on the left
Biggest change of all. The sky went dark and orange.



Sunset on the right
I asked for right, but the orange sky stayed on the left. A miss.

What still goes wrong here: the steel sculpture starts to look like stone, the pool loses its strong turquoise, and the wall tile lines get softer. So it's a nice picture, but not a true copy of the building.

Villa: first tries with each tool



Original photo. Goal: move the sun to the left.



Sun-Direction 9B
The sun moved, but the colors and pool went gray.



Qwen 3D Lighting
Soft haze on the left. It also zoomed the picture.



FLUX.1 Kontext
Almost nothing changed. The sun stayed where it was.



Sun-Direction 4B + color prompt
Colors came back. This led to the angle tests.

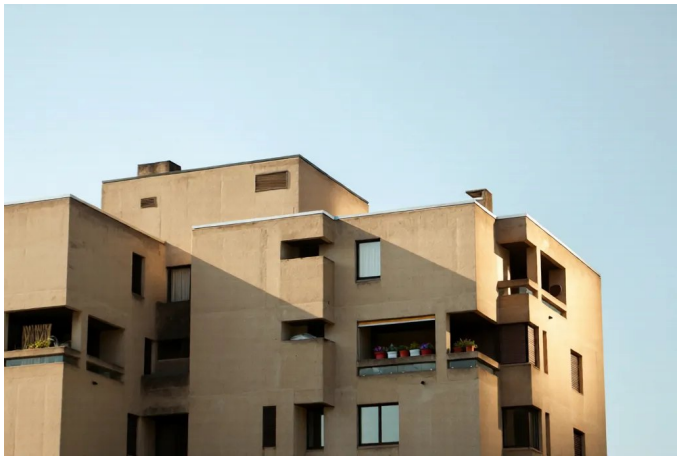
Concrete block



Original photo. Goal: move the sun to the right.



Sun-Direction 9B
Sun moved, but beige concrete turned gray.



Qwen 3D Lighting
Old shadow stayed. It zoomed out and added flower pots.



FLUX.1 Kontext
Diagonal shadow did not move.



Sun-Direction 4B + color prompt
Blue sky, moved sun, paler concrete. It added a tree.

Living room



Original photo. Goal: low sun from the back left, through the windows.



Sun-Direction 9B
Sofa, table, rug and the view outside all changed.



Qwen 3D Lighting
Furniture stayed, but it zoomed in and put sun on the window wall, which can't happen.



FLUX.1 Kontext
Nice sunset, but all orange, the view is new and the sun is on the wrong side.



Sun-Direction 4B + color prompt
Good light, but rug and armchair colors changed, plus an extra ceiling light.

Object check in the living room

This is the check that changed my mind. The overall score looked fine, but when I looked closely, the armchair lost its color, the table changed and the rug got smaller. Black areas are parts that fell outside that tool's frame.

Armchair


Original


Sun 9B


Sun 4B


Qwen


Kontext

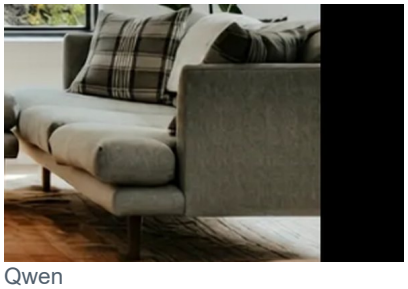

Light transfer

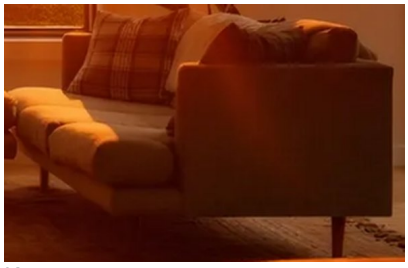
Sofa

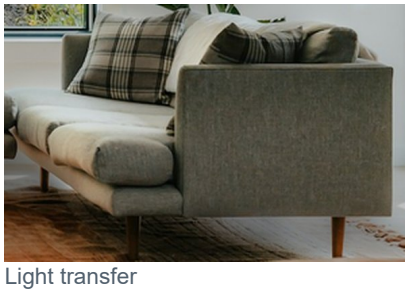

Original


Sun 9B

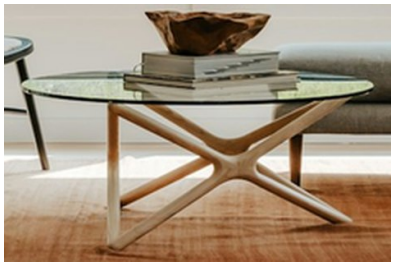

Sun 4B

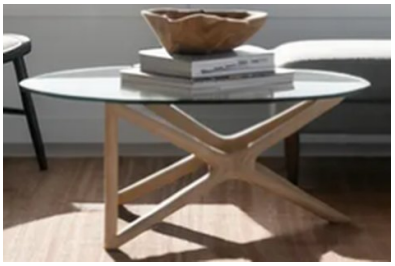

Qwen


Kontext


Light transfer

Coffee table

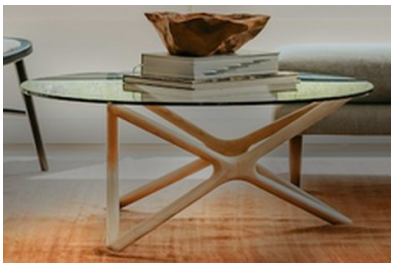

Original


Sun 9B


Sun 4B


Qwen


Kontext

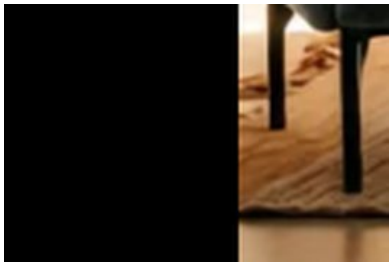

Light transfer

Rug, left end


Original


Sun 9B

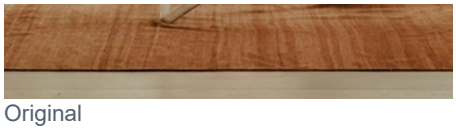

Sun 4B

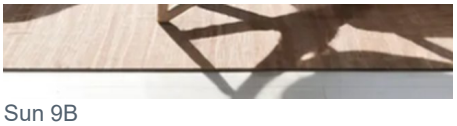

Qwen

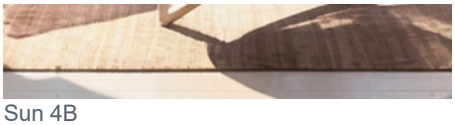

Kontext

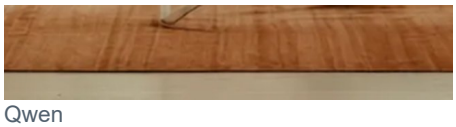

Light transfer

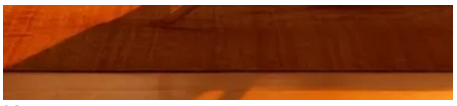
Rug, front edge

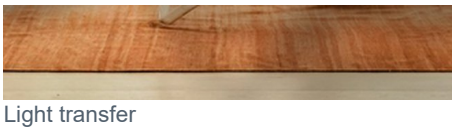

Original


Sun 9B


Sun 4B

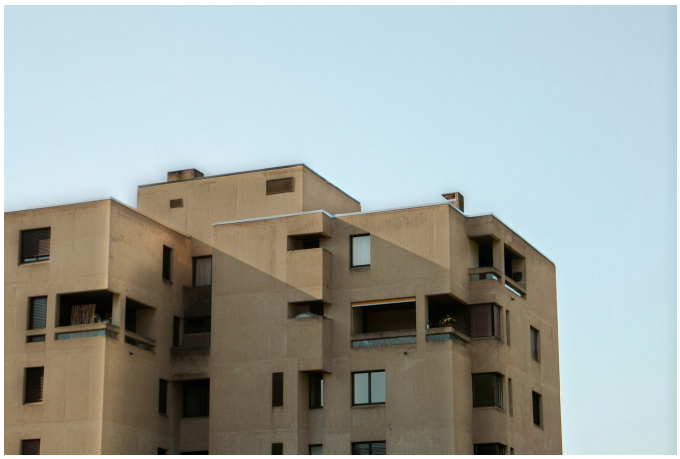

Qwen


Kontext

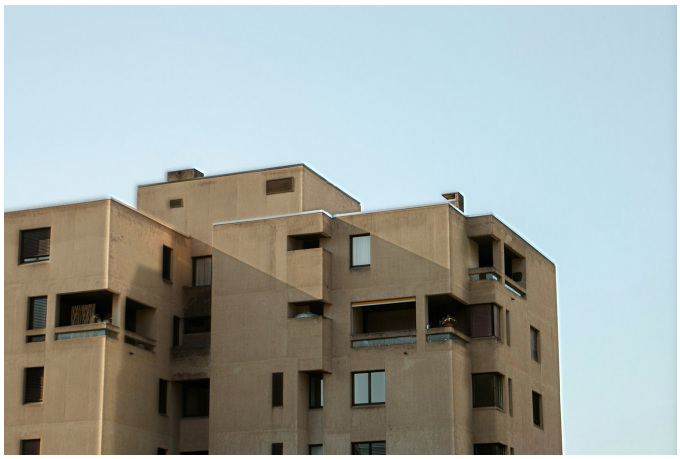

Light transfer

My own try: copy only the light

Since the models kept redrawing things, I kept the original photo and copied only the brightness change from a model's result. Objects stayed perfect, but the sun didn't really move. The old hard shadow on the concrete block is still there, and sharper settings added bright halos around edges.



Concrete, smooth setting
Objects are safe. The light barely changed.



Concrete, sharp setting
Old shadow stayed, with a bright halo on the roof.



Living room, smooth setting
Furniture is untouched. Soft light change.



Living room, sharp setting
More light, but halos and a ghost ring on the floor.

Errors I ran into

What happened	What I did
Several Hugging Face demos were paused or crashing	Found other Spaces that worked
The free GPU quota ran out after three images	Logged in to Hugging Face
Login looked fine, but the saved token was invalid	Found the problem and logged in again in the browser
Sun-Direction needed 150 s of quota and I had 147 s left	Moved the tests to my own GPU
The 9B model doesn't fit on my 12 GB card	Switched to the smaller 4B, which also has a better license
Sun-Direction needs a light ball drawn in the browser	Rewrote the ball in Python
Windows console crashed on special characters	Forced UTF-8 output in the scripts
My first score went to almost zero when a model zoomed	Added automatic alignment before scoring
Kontext got the best score for barely changing anything	Stopped trusting one number, and scored the light separately
The score missed the changed furniture	Added the object check
I thought the gray colors came from the overcast step	Saved that step and saw it was the LoRA pass. A color prompt helped
Villa left and right sun looked the same	Used a brighter ball, a stronger prompt and a lower sun
Sunset on the right still glowed on the left	Left it in as a failure. Sunset direction isn't reliable yet
Light transfer left the old shadows in place	Kept it as a safety idea, not the final answer

What I would put in production

I'd run the Sun-Direction 4B LoRA on a serverless GPU that only runs when someone uses it, and offer it for exteriors only. After generating, an automatic check compares the result to the original and tries again if the building changed.

Option	Cost per image	Speed	My take
Own serverless GPU (RTX 4090)	About $0.01, or $0.02 with a retry	About 20 to 26 s	My pick
Always-on GPU	About $650 a month	Same, no cold start	Only above about 16,000 images a month
Hosted Kontext or Qwen API	$0.02 to $0.05	Not measured	Open versions failed this task

These cost and speed numbers are estimates from RunPod prices and my 9B Space times. I did not test 4B on that GPU.

What could make it better

- **Test more photos:** try at least 30 exteriors and see how often a result passes. I only tried three photos, each with one seed.
- **Protect the materials:** mask glass, metal and water, so that the model can't turn steel into stone.
- **Use depth:** get a depth map of the photo and calculate real shadows from the true sun position.
- **Mix both worlds:** combine the model's new shadows with the original textures, so we get real light and real materials.
- **Train for interiors:** a LoRA trained on rooms, or use masks to lock the furniture.
- **Make it faster:** test a bigger GPU, since my card was short on memory. Your API averages 13.4 s, and mine is slower.
- **Add a real sun control:** let the user drag the sun on a dome, with the date, time and location of the building.
- **Test renders too:** I only used real photos here, but your users upload renders.

All code, images and failed tries are on GitHub: https://github.com/mehmetztrk/sun-direction-relighting

Photos from Unsplash: framesforyourheart, Clay Banks, Valentin Gerardi.